COMPARING THE TWO DEFINITIONS OF TOR INDEPENDENCE OVER A FLAT ALGEBRA

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tor independence of a pair of modules over a ring. defined in the same section, source lines 15000-15037, before the counted statement. Supplies both sides of the comparison.
- A module can be represented by a flat or free resolution when computing derived tensor products. used at proof lines 15049-15070 to replace the derived constructions by total complexes. Makes the two sides concrete objects that can be compared.
- If the algebra is flat, tensoring with it preserves quasi-isomorphisms and commutes with the totalisation. used at proof lines 15070-15099. Is the hypothesis that lets the comparison be carried out by reordering tensor factors.

Source reference. Original lemma 0DJD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16973–16973. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10794–10794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Assumptions as in Lemma 08HW. For $M \in D(A)$ there are canonical isomorphisms*

$$H^i((M \otimes^{\mathbf{L}}_A A') \otimes^{\mathbf{L}}_{R'} B') = H^i(M \otimes^{\mathbf{L}}_R B) \otimes_{(A \otimes_R B)} (A' \otimes_{R'} B')$$

of $A' \otimes_{R'} B'$-modules.

Proof. Let us elucidate the two sides of the equation. On the left hand side we have the composition of the functors $D(A) \to D(A') \to D(R') \to D(B')$ with the functor $H^i : D(B') \to \mathrm{Mod}_{B'}$. Since there is a map from A' to the endomorphisms of the object $(M \otimes^{\mathbf{L}}_A A') \otimes^{\mathbf{L}}_{R'} B'$ in $D(B')$, we see that the left hand side is indeed an $A' \otimes_{R'} B'$-module. By the same arguments we see that $H^i(M \otimes^{\mathbf{L}}_R B)$ has an $A \otimes_R B$-module structure.

We first prove the result in case $B' = R' \otimes_R B$. In this case we choose a resolution $F^\bullet \to B$ by free R-modules. We also choose a K-flat complex $M^\bullet$ of A-modules representing M. Then the left hand side is represented by

$$\begin{aligned} H^i(\mathrm{Tot}((M^\bullet \otimes_A A') \otimes_{R'} (R' \otimes_R F^\bullet))) &= H^i(\mathrm{Tot}(M^\bullet \otimes_A A' \otimes_R F^\bullet)) \\ &= H^i(\mathrm{Tot}(M^\bullet \otimes_R F^\bullet) \otimes_A A') \\ &= H^i(M \otimes^{\mathbf{L}}_R B) \otimes_A A' \end{aligned}$$

The final equality because $A \to A'$ is flat. The final module is the desired module because $A' \otimes_{R'} B' = A' \otimes_R B$ since we've assumed $B' = R' \otimes_R B$ in this paragraph.

General case. Suppose that $B' \to B''$ is a flat ring map. Then it is easy to see that

$$H^i((M \otimes^{\mathbf{L}}_A A') \otimes^{\mathbf{L}}_{R'} B'') = H^i((M \otimes^{\mathbf{L}}_A A') \otimes^{\mathbf{L}}_{R'} B') \otimes_{B'} B''$$

and

$$H^i(M \otimes^{\mathbf{L}}_R B) \otimes_{(A \otimes_R B)} (A' \otimes_{R'} B'') = \big(H^i(M \otimes^{\mathbf{L}}_R B) \otimes_{(A \otimes_R B)} (A' \otimes_{R'} B')\big) \otimes_{B'} B''$$

Thus the result for B' implies the result for B''. Since we've proven the result for $R' \otimes_R B$ in the previous paragraph, this implies the result in general. □